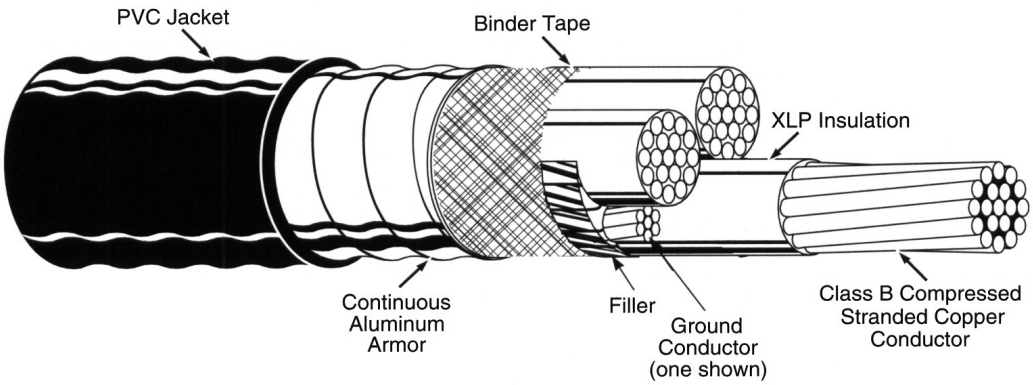
HLX02XXH

ARMOR-X and ARMOR-XTRA. -HL. 600 Volt Type MC-HL Power. XLP Insulated, 90°C. Continuous Corrugated Aluminum Armor with -40°C Jacket.



APPLICATIONS

Southwire's 600 Volt HLX02XXH ARMOR-X and ARMOR-XTRA Type MC-HL Power Cable provides an impervious sheath recommended as an economical alternative to traditional conduit systems. For use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs or continuous rigid cable supports. For use in Classes I, II and III, Division 1 and 2, hazardous locations covered under NEC Articles 501, 502 and 503. The cable is designed with three symmetrically placed grounds to reduce problems associated with pulse-width modulated ac drives. These cables are capable of operating continuously in wet or dry locations at a maximum conductor temperature of 90 ^{O} C for normal operation, 130 ^{O} C for emergency overload conditions, and 250 ^{O} C for short circuit conditions. Maximum sidewall pressure is 1000 lbs.

STANDARDS

Southwire's 600 Volt HLX02XXH ARMOR-X and ARMOR-XTRA Type MC-HL Power Cable is manufactured and tested in accordance with the latest revisions of:

- UL 44 - Thermoset-Insulated Wires and Cables.
- UL 1569 - Metal-Clad Cables.
- UL 1309 - Listed as Marine Shipboard Cable.
- UL 2225 - Safety Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified) Locations.
- ABS Listed as CWCMC.
- IEEE 1580 - Incidental Motion "IM" Compliance.
- IEEE 1202 - Flame Testing of Cables for Use in Cable Tray in Industrial and Commercial Occupancies (70,000 Btu/hr).
- IEC 60332-3-22
- IEC 60092-350
- ICEA T-29-520 - Vertical Cable Tray Flame Tests (210,000 Btu/hr).
- ICEA S-58-679 - Control Cable Conductor Identification.
- CSA C.22.2 No.123 & 239
- Canadian Electrical Code Part 1.12-700.

Southwire's 600 Volt HLX02XXH ARMOR-X and ARMOR-XTRA Type MC-HL Power Cable meets and exceeds all construction requirements of ICEA S-95-658 (NEMA WC 70) - Nonshielded 0 - 2 kV Cables, with testing frequencies based on UL requirements.

CONSTRUCTIONS

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

MADE IN THE USA

Copyright 2003, Southwire Company. All Rights Reserved.
®Southwire is a registered trademark of Southwire Company.

HLX02XXH 3c/4c XHHW Power, 3xGW, Cont. Alum. Weld. Armor,

Southwire's 600 Volt HLX02XXH ARMOR-X and ARMOR-XTRA Type MC-HL continuous corrugated aluminum armor provides an aluminum sheath impervious to moisture, liquids and gases. Cable is flame retardant UL-listed for cable tray use, direct burial, sunlight resistant and minus 40°C low temperature rating. Continuous aluminum sheath meets the grounding requirements of NEC 250.122 in non-HL locations (Class I Div I). Conductors are UL-listed Type XHHW-2 with phase id.

Product Code Three Conductor	Size	Conductor Diameter		Insulation Thickness		Grounds	Approx. Core Diameter		Approx. ARMOR-X Diameter		Jacket Thickness		Approx. Overall Diameter		Approx. Net Weight		Allowable Ampacity
	AWG/ kcmil	inch	mm	inch	mm	No. x AWG	inch	mm	inch	mm	inch	mm	inch	mm	lb./Mft.	kg/km	90°C
HLX01-XXH-14/3/P	14	.07	1.77	.03	.76	3 X 18	.29	7.37	.5	12.6	.05	1.27	.6	15.1	175	260	25
HLX01-XXH-12/3/P	12	.087	2.22	.03	.76	3 X 16	.327	8.31	.55	13.8	.05	1.27	.65	16.4	221	328	30
HLX01-XXH-10/3/P	10	.111	2.81	.03	.76	3 X 14	.367	9.33	.58	14.6	.05	1.27	.68	17.1	283	421	40
HLX01-XXH-08/3	8	.139	3.53	.045	1.14	3 X 14	.527	13.39	.75	18.9	.05	1.27	.85	21.5	405	602	55
HLX01-XXH-06/3	6	.174	4.43	.045	1.14	3 X 12	.572	14.53	.8	20.2	.05	1.27	.9	22.7	541	805	75
HLX01-XXH-04/3	4	.221	5.6	.045	1.14	3 X 12	.647	16.44	.91	23.1	.05	1.27	1.01	25.7	698	1039	95
HLX01-XXH-02/3	2	.277	7.04	.045	1.14	3 X 10	.807	20.5	1.05	26.7	.05	1.27	1.15	29.2	1043	1552	130
HLX01-XXH-010/3	1/0	.362	9.19	.055	1.4	3 X 10	1.06	26.92	1.35	34.3	.05	1.27	1.45	36.8	1675	2492	170
HLX01-XXH-020/3	2/0	.405	10.29	.055	1.4	3 X 10	1.18	29.97	1.47	37.3	.05	1.27	1.57	39.9	1857	2763	195
HLX01-XXH-030/3	3/0	.454	11.53	.055	1.4	3 X 8	1.23	31.24	1.54	39.1	.06	1.52	1.66	42.16	2344	3488	225
HLX01-XXH-040/3	4/0	.51	12.95	.055	1.4	3 X 8	1.38	35.05	1.67	42.4	.06	1.52	1.79	45.5	2792	4155	260
HLX01-XXH-250/3	250	.558	14.17	.065	1.65	3 X 8	1.55	39.37	1.87	47.5	.06	1.52	1.99	50.5	3277	4876	290
HLX01-XXH-350/3	350	.661	16.79	.065	1.65	3 X 6	1.79	45.47	2.2	55.9	.06	1.52	2.32	58.9	4420	6577	350
HLX01-XXH-500/3	500	.79	20.07	.065	1.65	3 X 6	2.028	51.51	2.43	61.7	.075	1.91	2.58	65.5	6001	8929	430
HLX01-XXH-750/3	750	.968	24.59	.08	2.03	3 X 4	2.576	65.43	2.98	75.7	.075	1.91	3.13	79.5	9375	13949	535

Ampacities shown are for general use as specified by the NEC, 2008 Edition, Section 310.15, Table 310.16 90°C. Ampacities for more than three current-carrying conductors are adjusted as required by Notes to Ampacity Tables of 0 to 2000 volts.

Product Code Four Conductor	Size	Conductor Diameter		Insulation Thickness		Grounds	Approx. Core Diameter		Approx. ARMOR-X Diameter		Jacket Thickness		Approx. Overall Diameter		Approx. Net Weight		Allowable Ampacity
	AWG/ kcmil	inch	mm	inch	mm	No. x AWG	inch	mm	inch	mm	inch	mm	inch	mm	lb./Mft.	kg/km	90°C
HLX01-XXH-14/4/P	14	.07	1.77	.03	.76	3 x 18	.327	8.31	.55	13.8	.05	1.27	.65	16.4	202	301	20
HLX01-XXH-12/4/P	12	.087	2.22	.03	.76	3 x 16	.367	9.33	.58	14.6	.05	1.27	.68	17.1	252	376	24
HLX01-XXH-10/4/P	10	.111	2.81	.03	.76	3 x 14	.405	10.29	.63	15.9	.05	1.27	.73	18.4	332	494	32
HLX01-XXH-08/4	8	.139	3.53	.045	1.14	1 x 10	.572	14.53	.8	20.2	.05	1.27	.9	22.7	465	692	44
HLX01-XXH-06/4	6	.174	4.43	.045	1.14	1 x 8	.647	16.44	.91	23.1	.05	1.27	1.01	25.7	640	952	60
HLX01-XXH-04/4	4	.221	5.6	.045	1.14	1 x 8	.744	18.9	1.01	25.7	.05	1.27	1.11	28.2	885	1317	76
HLX01-XXH-02/4	2	.277	7.04	.045	1.14	1 x 6	.897	22.78	1.22	31	.05	1.27	1.32	33.5	1291	1921	104
HLX01-XXH-010/4	1/0	.362	9.19	.055	1.4	1 x 6	1.18	29.97	1.47	37.3	.05	1.27	1.57	39.9	1961	2917	136
HLX01-XXH-020/4	2/0	.405	10.29	.055	1.4	1 x 4	1.31	33.27	1.59	40.4	.06	1.52	1.71	43.4	2376	3535	156
HLX01-XXH-040/4	4/0	.51	12.95	.055	1.4	1 x 4	1.55	39.37	1.87	47.5	.06	1.52	1.99	50.5	3544	5273	208
HLX01-XXH-250/4	250	.558	14.17	.065	1.65	1 x 4	1.68	42.67	2.04	51.8	.06	1.52	2.16	54.9	4052	6029	232

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

MADE IN ★ THE USA

Copyright 2003, Southwire Company.
All Rights Reserved.

®Southwire is a registered trademark of Southwire Company.

Armor-X, HL

HLX01-XXH-350/4	350	.661	16.79	.065	1.65	1 x 3	1.96	49.78	2.29	58.2	.075	1.91	2.44	62	5602	8336	280
HLX01-XXH-500/4	500	.79	20.07	.065	1.65	1 x 2	2.232	56.69	2.67	67.8	.075	1.91	2.82	71.6	7721	11489	344
HLX01-XXH-750/4	750	.968	24.59	.08	2.03	1 x 1	2.753	69.93	3.22	81.8	.085	2.16	3.39	86.1	11964	17803	428

Ampacities shown are for general use as specified by the NEC, 2008 Edition, Section 310.15, Table 310.16 90°C. Ampacities for more than three current-carrying conductors are adjusted as required by Notes to Ampacity Tables of 0 to 2000 volts.

SCOPE: This specification covers three or four conductor XLP (cross-linked polyethylene) insulated, continuous corrugated aluminum armored, thermoplastic jacketed, 600-Volt cable with grounding conductor for use in aerial installations, metal racks, cable trays, troughs or continuous rigid cable supports. The cable will be listed Type MC-HL for Classes I, II and III, Division 1 and 2, hazardous locations and have three symmetrically placed grounds to reduce problems associated with pulse-width modulated ac drives. This cable is capable of operation continuously at a conductor temperature of 90°C in wet or dry locations.

STANDARDS: The following standards will form part of this specification - ASTM B8, ICEA S-95-658/NEMA WC70, UL 44 and UL 1569.

CONDUCTOR: The conductor will be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8 and ICEA.

INSULATION: The insulation will be XLP meeting the requirements of the referenced standards. The insulation thickness will be listed in ICEA, and the minimum spot thickness will not be less than 90% of the listed amounts. Individual conductors will be color coded per ICEA S-58-679, using Method 3, Table 2 for sizes AWG 4 and larger, and using Method 1, Table 2 for sizes AWG 6 and smaller.

ASSEMBLY: The insulated conductors will be cabled round with three symmetrically placed ground wires, fillers and covered with a binder tape.

ARMOR: Continuous corrugated copper-free aluminum sheath shall be applied over the assembly. The continuous sheath will be impervious to moisture, liquids and gases.

JACKET: The cable will be covered with a black PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness will be in accordance with ICEA, and the minimum spot thickness will not be less than 80% of the average thickness. The jacket will be sunlight resistant and will meet the requirements of the IEEE 1202 (70,000 Btu/hr) and ICEA T-29-520 (210,000 Btu/hr) vertical cable tray flame tests. The jacket will be suitable for use at a minimum ambient temperature of -40°C.

IDENTIFICATION: Manufacturer's identification will be printed on the jacket.

TESTS: Physical and electrical tests will be conducted in accordance with the requirements of the referenced standards.

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

MADE IN ★ THE USA

Copyright 2003, Southwire Company.
All Rights Reserved.

®Southwire is a registered trademark of Southwire Company.